

Disproof of Conjecture 00000003711

Resistance sums and spanning trees of a triangle

gaochengzhecpu

Claim addressed. The conjecture identifies the sum of the resistance-distance quadratic forms with the number of spanning trees. The connected simple graph K_3 , with unit conductance on each edge, disproves this identification under both unordered-pair and ordered-pair conventions. The individual pseudoinverse formula for resistance is not disputed.

Proposition 1. *For the unit triangle, every pair of distinct vertices has effective resistance $2/3$. The sum over unordered vertex pairs is 2, the sum over ordered pairs is 4, and the number of spanning trees is 3.*

Proof. Let I be the 3×3 identity and J the all-ones matrix. The graph Laplacian and its Moore–Penrose inverse are

$$L = 3I - J = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}, \quad Q = \frac{1}{3}I - \frac{1}{9}J.$$

Indeed $J^2 = 3J$, so $LQ = QL = I - J/3$. Direct multiplication gives $LQL = L$, $QLQ = Q$, and both LQ and QL are symmetric. These are the four defining Moore–Penrose equations.

For vertices i, j , write $b = e_i - e_j$. Then $Jb = 0$, and hence

$$v = Qb = \frac{1}{3}b, \quad Lv = b.$$

This is the voltage solving Kirchhoff’s law for a unit current entering at i and leaving at j . Its terminal drop equals the quadratic form:

$$R_{ij} = v_i - v_j = b^T Qb = \begin{cases} 0 & i = j, \\ 2/3 & i \neq j. \end{cases}$$

The drop does not depend on the voltage representative. If $Lw = 0$, subtracting the first two equations gives $w_0 = w_1$, and either equation then gives $w_2 = w_0$. Thus two solutions differ by a constant.

There are three unordered pairs and six ordered distinct pairs. Their sums are consequently $3(2/3) = 2$ and $6(2/3) = 4$. A spanning tree of K_3 consists of exactly two of its three edges. Each such choice is connected and acyclic, and there are exactly three choices. Therefore neither resistance sum equals the spanning-tree count 3. $\square$

Scope. This is a connected, unweighted, loopless simple graph; no missing connectedness hypothesis is exploited. Including diagonal pairs adds zero. The source’s assertion about the sum fails with either usual pair-counting convention. No assertion is made against the usual matrix-tree theorem or the individual effective-resistance formula.

Lean correspondence and reproduction

The project defines K_3 as Mathlib’s complete `SimpleGraph` (Fin 3). Its actual `lapMatrix` is proved equal to L . The four Moore–Penrose equations for the real matrix Q are proved entry by entry. The current, voltage and quadratic form are genuine matrix-vector and dot products. In addition, `electrical_resistance` proves that every solution of the actual Kirchhoff equation has the claimed terminal drop.

Spanning trees are subgraphs on the full vertex set satisfying Mathlib’s actual `IsTree`. They are placed in bijection with the three-edge masks having exactly two edges. The proof uses the connectedness and edge-count characterization of finite trees, with finite connectedness checked by the library’s walk-based decision procedure. Thus the number 3 is a derived cardinality, not a count assigned by definition.

The final theorems `unordered_conjecture_false` and `ordered_conjecture_false` compare the actual quadratic-form sums with that cardinality.

Inside `lean/`, run `lake build`, followed by

```
lake env lean -DwarningAsError=true Main.lean.
```

Lean is pinned to 4.19.0 and Mathlib to

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

The dependency manifest pins every transitive dependency. The supplementary `verify.py` independently checks the matrix equations and tree enumeration with exact rational arithmetic. There are no proof gaps, custom axioms or native computation shortcuts. Actual fresh-build logs, axiom dependencies and artifact hashes are in `verification/`. Author and delegated-agent review records are separate; no external independent review is claimed.

References

- [1] The Last Math Competition, Conjecture 00000003711 (original bilingual source).